YCCify - Yeshwantrao Chavan College of Engineering

IT701 - Machine Learning | Dr. S. S. Khedkar

Unit 5 - Neural Networks & Deep Learning

Topic: Perceptrons & Backpropagation

5.1 Artificial Neural Networks

Biologically inspired computational graphs.

Foundations:

* Single-layer Perceptron
* Multilayer Perceptron MLP
* Activation Functions: Sigmoid, Tanh, ReLU, LeakyReLU
* Forward Propagation and Loss Functions Cross Entropy, MSE

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5.2 Backpropagation Algorithm

Gradient descent optimization through chain rule differentiation.

Optimization Algorithms:

* Stochastic Gradient Descent SGD
* Adam Optimizer with Momentum
* Regularization: Dropout, Batch Normalization